The Roles of the HCT Nurse

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Michelle Kenyon and John Murray

32.1 Introduction: HCT Nursing

Haematopoietic cell transplantation (HCT) is undoubtedly one of the most challenging and complex forms of treatment for malignant and non-malignant blood disorders as well as autoimmune disease. As a result, nursing in the field of HCT and cellular therapy offers a wealth of opportunities to make a positive difference to patient experience through high-quality specialist nursing care and the unique role of nursing within the transplant MDT. In this section, we will describe the nursing roles that surround the patient pathway and offer an overview the particular aspects of patient care that they encompass.

More than 82,000 transplants are performed globally each year (Niederwieser et al., 2021), and this is increasing, leading to ongoing workforce demands on the background of a largescale shortage of qualified nurses. While global regions face their own specific challenges, poor pay and working conditions for many nurses globally means sustained commitment by governments and policymakers are needed to urgently address these issues (Jester 2023). Our transplant units, where it takes time to train and develop new staff, are vulnerable to the current workforce deficit and while recruitment is challenging in the face of a global shortage, we can do a lot to retain the staff that we do have. Well-defined career pathways together with a professional development strategy and specialist training opportunities can be a very effective investment.

There is great potential to improving the care that we provide to patients and opportunities for nursing research and innovation should be optimised through local and national research networks. In a recent review of cancer nursing research priorities (Dowling et al., 2023), the role of technology in improving patient and caregiver symptoms and health outcomes was top. Other priorities included those focused on culturally sensitive psychosocial care, financial toxicity, social determinants of health and scope of practice; all highly relevant to the transplant and cell therapy sphere.

In this continually evolving speciality, where nurses are pivotal to patient outcomes, the importance of a well-equipped, professionally educated, knowledgeable and competent workforce is critical. This is reinforced in the FACT-JACIE International Standards where accreditation requires that the clinical programme has access to personnel who are formally trained, experienced and competent in the management of patients receiving cellular therapy (JACIE 8th

M. Kenyon (✉)
Department of Haematological Medicine, King's College Hospital NHS Foundation Trust, London, UK
e-mail: michelle.kenyon@nhs.net

J. Murray
Haematology and Transplant Unit, Christie Hospital NHS Foundation Trust, Manchester, UK
e-mail: j.murray10@nhs.net

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edition standards 2021). Therefore, transplant centres must commit to ongoing nurse education, professional development and research in all aspects spanning novice to expert to ensure sustainability and capability to deliver complex, state-of-the-art treatments in the years ahead.

32.2 The Roles of Nursing Throughout the Patient Transplant and Cell Therapy Pathway

Nursing is recognised as both an emerging profession and an academic discipline. The diverse and continually evolving field of transplantation and cell therapy creates many opportunities for innovation in nursing roles and as such these have been developed to support the patient pre-, peri-, and post-treatment. Nurses and in particular the Clinical Nurse Specialist (CNS) are evaluated as having a positive impact on service delivery (Kerr et al., 2021) in relation to continuity of care, patient advocacy and increased access to services such as the medical consultant (Beaver et al. 2010, Ream et al. 2009). Regardless of the stage of pathway that nurses deliver care to patients, they should be aware of the complications in order to play a role in prevention or early detection of illness such as sepsis, dehydration, electrolyte imbalance and organ dysfunction, taking appropriate measures to minimise adverse effects and deliver prescribed treatment. This care is complex and requires a high level of skill (Walhult et al. 2023).

32.2.1 (Central) Venous Access Devices (cVADs)

All nurses, regardless of role, require education and training on the care and maintenance of cVADs, and this should include patient well-being and safety. There are a range of devices, and device selection should be based on considerations such as diagnosis, proposed treatment and vein condition.

Within the available devices, a PICC (peripherally inserted central catheter) is frequently used, especially in autologous transplant and CAR-T, with a tunnelled catheter such as a Hickman often reserved for allogeneic transplant particularly those receiving myeloablative conditioning therapy.

Nurses are responsible for the care of cVADs and are the main users of these devices for infusions of chemotherapy, medication, parenteral nutrition and transfusions. Safe handling and care of the cVAD and infusions are vital in this process due to the risks associated with catheter-related infections. Aseptic Non-Touch Technique (ANTT) (Pratt et al. 2007, Rowley et al. 2010) has led to a decrease in catheter-related infections.

32.2.2 GvHD

Graft-versus-host disease (GvHD) is the most recognised complication post-HCT and was first observed in 1956 (Ghimire et al. 2017) and remains a leading cause of non-relapse mortality and is associated with a high morbidity that increasingly affects quality of life (Lee et al. 2002). Nursing care of patients with GvHD is highly complex and extremely stressful especially in the acute setting in patients with grades 3–4 skin and GI involvement. Supportive nursing care to complement medical interventions aims to offer symptomatic comfort and relief. There are many manifestations of GvHD, and nurses are able to advise patients with respect to many of these including eye, mouth and genital care. For further readings refer to the GvHD chapter in The EBMT Textbook for Nurses (Kenyon & Babic, 2023).

32.3 The Transplant Coordinator

The transplant coordinator (TC) is responsible for the preparation of patients and donors in a complex process that requires an expert level of communication and planning. Each patient and their families need to be physically and psycho-

logically prepared to enable them to manage the treatment; the TC is at the centre of this process. Many transplant coordinators are nurse specialists who focus their role on the individual needs of the patient and families; however, some centres have medical staff that occupy this role. Transplant coordinators provide a high level of care and management, with explanations of complicated and complex tests and many have a wide area of latitude to make clinical decisions within departmental SOP's. The TC will participate in specific or advanced nursing practices, e.g. bone marrow sampling, HLA typing, and transplant recipient care.

The TC ensures that a suitable source of cells (autologous or allogeneic) is available. This entails requests to donor search panels and ordering cells once an ideal match has been identified by the transplant physician. The TC supports the patient with verbal and written information and educates them about the whole process. The TC will coordinate all of the care and embodies a clinical nursing function where emphasis is placed on specialisation in a clearly defined area of care.

The TC may also in some institutions take care of the donor, arranging tissue typing and will liaise with the apheresis or bone marrow harvesting team to ensure a smooth process. Sibling donors may often have additional anxieties to that of an unrelated donor, and support from an experienced TC is vital. The TC is involved in the creation of information tools for the patient and the donor which are evaluated in order to have an accurate knowledge of patients' needs. The TC actively participates in the JACIE process of accreditation of transplant centres by writing and evaluating SOPs and being a valued member of the MDT and ward team offering teaching and advice (Kenyon, Babic 2023).

32.4 The Apheresis Nurse

The apheresis nurse is a highly skilled practitioner, who is able to handle the technicalities of harvesting as well as the social, emotional and ethical aspects (Neyrink & Vrielink, 2019) and be adept at managing an often-fluid starting date and time. Apheresis nurses are an integral part of the transplant process, and the work-up begins as soon as a patient or donor is identified by the team as suitable for harvest. The process for patients and donors (siblings or unrelated) is similar. There are routine blood tests and screening to be performed that are driven by the specific protocol which must be in place prior to the harvest commencing. Once a proposed harvest date is set, for those undergoing an autologous procedure the conditioning therapy can be commenced. For donors, a date for the first dose of G-CSF is given. The day of harvest may change in real time as blood counts may not be optimal on the first day of attendance. This requires excellent administrative and communication skills to rearrange the procedure and inform all concerned parties at short notice of the delay. Often more than 1 day of apheresis is required which again may lead to logistical challenges. Especially if an unrelated donor and cells are to be collected to be taken elsewhere.

The apheresis nurse spends a significant amount of time with the patient or donor during the day(s) of harvest and is able to provide information and answer questions about the transplant process. This can often be a nervous time for the patient who is aiming to provide enough cells for a transplant (HCT or CAR-T) in the future or for the donor who is hoping to give enough cells for a successful transplant for a relative or unrelated recipient. An apheresis nurse is not just a technician.

32.5 The HCT Ward Nurse

Nurses, in particular those caring for patients during the peri-transplant period and the early months after engraftment, are pivotal in implementing practices to prevent and manage infections and other serious effects following HCT (Kenyon & Babic, 2023) such as:

- Bleeding caused by thrombocytopenia

- Fatigue caused by decreased haemoglobin levels and the effects of chemo/radiotherapy and associated medication
- Oral hygiene and pain management due to mucositis
- Gastrointestinal toxicity including nausea, vomiting, diarrhoea and constipation
- Sepsis
- Impaired nutritional intake, weight loss and malnutrition
- Psychosocial concerns
- Effects of protective isolation

The ward nurse, working at the bedside with the patient, is best placed to monitor for early and acute complications. Early complications are generally considered to be those that occur during the first 3 months following transplantation when the patient has reduced tolerance due to neutropenia and/or increased intestinal permeability and in the allogeneic setting, high dose immunosuppression. In neutropenia, the number of white blood cells decreases significantly, resulting in increased risk of infection. An increased permeability of the intestinal wall caused by intensive chemotherapy damages the gastrointestinal mucosa. As a result, pathogenic bacteria (bodily bacteria or bacteria from the diet) can enter the bloodstream and cause sepsis.

In the early phase of HCT, the main risk factors for infections are neutropenia-barrier breakdown due to mucositis, indwelling catheters, depressed T-cell and B-cell function and aGvHD. Two of the most common early complications, **oral mucositis and sepsis,** will be discussed below. Other complications such as haemorrhagic cystitis (HC), eosinophilic syndrome (ES) and diffuse alveolar haemorrhage (DAH) occur less often but can be serious when they do arise. Transplant-associated thrombotic microangiopathy (TA-TMA) and veno-occlusive disease (VOD) are analysed in Chaps. 42 and 49. For all complications, there are locally agreed SOPs often supported by national guidelines, recommendations for prevention and principles for nursing care, with monitoring and prompt intervention that can influence patients' morbidity and mortality.

32.5.1 Oral Mucositis (OM)

Oral mucositis (OM) is the inflammation of the mucosal membrane, characterised by ulceration, which may result in pain, swallowing difficulties and impairment of the ability to talk (Al-Dasoogi et al., 2013). The mucosal injury offers a ground for infection which can potentially lead to sepsis and septicaemia (Quinn et al., 2020). However, OM is not the only oral complication seen within the transplant setting but most patients undergoing autologous and allogeneic HCT will experience mucosal changes within their oral cavity leading to difficulties in eating, sleeping and talking and a reduction in quality of life.

Care strategies are aimed at optimising care of the oral cavity, preventing oral damage, infection prevention and treatment of oral complications when they arise. More information on oral mucositis, refer to the early and acute complications chapter in The EBMT Textbook for Nurses (2023).

32.5.2 Sepsis

Patients undergoing HCT are at increased risk of infection, and this is a leading cause of morbidity and mortality. The signs and symptoms of sepsis can be subtle and sometimes difficult to recognise in the presence of neutropenia and other transplant complications. Preventive measures are important but increased monitoring and the use of early warning scores alongside team collaboration and immediate action can be life-saving, allowing for prompt and appropriate sepsis management.

In the early phase of transplant, the main risk factors for infection are (Rovira et al., 2012)

- Neutropenia
- Barrier breakdown
- Depressed T- and B-cell function
- Prescence of aGvHD

Responsibility for the implementation of strategies for infection prevention and control extends to the whole HCT MDT and includes:

- Hand hygiene
- Respiratory hygiene
- PPE
- Safe management and care of equipment
- Safe management of the environment
- Management of laundry
- Management of blood and body fluid spills
- Waste management
- Management of exposure

Early recognition and treatment are vital for a successful outcome of sepsis. Temperature, pulse, blood pressure, respirations and saturation (vital signs) should be frequently monitored. Signs of infection are not always obvious, but if the patient has a temperature ≥38.0 °C, cultures should be taken, IV antibiotics and IV fluids started or increased and oxygen therapy initiated. The goal is always to *start antibiotic treatment within 1 h* from detection of fever (Swedish "Pro Sepsis" Programme Group Sepsis 2015). This is sometimes referred to as "the golden hour" (or "door-to-needle time" for patients admitted from outside the hospital) and is the most critical period in the patient's survival from sepsis.

The concept of the *Sepsis Six* has been developed as a guide to prioritise interventions and offer a resuscitation bundle in patients where sepsis is suspected (Daniels et al. 2011).

1. Oxygen therapy
2. Blood cultures
3. IV antibiotics
4. Fluid resuscitation
5. Serum lactate
6. Assess urine output (may require catheterisation)

Patients with sepsis are likely needed additional nursing care such as assistance with oral care and personal hygiene. It is important to ensure that the patient's and caregivers' information, education and support needs are met.

On discharge from the hospital, we need to ensure that the patient and their caregiver are aware of when, why and how to contact the clinic or hospital that they have a fever thermometer at home, know when to take their temperature and are aware of the level that constitutes a fever. For further information on the nursing care of the septic patient refer to the EBMT Textbook for Nurses (2023).

Since transplantation is a complex treatment with significant risk for some patients, a previously curative intent may evolve to end of life care. The nurse is a key advocate for their patients and should have the opportunity to work with the MDT to ensure that the patient's wishes and best interests are taken into consideration when a positive treatment outcome no longer be possible.

32.6 Advanced Clinical Practice Nursing in HCT

The advanced clinical practice (ACP) nurse role in HCT has rapidly developed over the past 20 years in Europe. The ACP may be ward- or outpatient-based, will usually have prescribing, admission and radiology rights and will diagnose and manage patients alongside medical colleagues. The aim is to improve patient care through medication management, patient and staff education, implementation of protocols and guidelines and developing quality improvement initiatives to improve outcomes (Mahmoudjafari et al. 2023). This position is embedded in a multi-professional framework for advanced clinical practice developed in 2017 in England (Multi-professional framework for advanced clinical practice in England—Advanced Practice (hee.nhs.uk) 2017), and similar documentation exists across Europe that allows for new, dynamic and flexible ways of working and delivering high quality care. There are capabilities that underpin ACP such as clinical practice, leadership and management, education and research that if applied to this post ensure a quality service is provided. The ACP as an experienced member of the transplant team bridges nursing and medical care to enhance patient experience.

32.7 Post-HCT Nursing

Embedding SOPs and protocols within transplant care has enabled nurses to develop important roles in the follow-up period after HCT. Nurses are an important source of support in the immediate period after patients go home. Even though patients are given information during their transplant and prior to discharge to prepare them for this stage of recovery, this is a time of great uncertainty and change. In addition, patients remain clinically vulnerable to the development of early complications and regular HCP contact and support are critical to outcomes. The nurse involved in post-HCT care requires specific training on monitoring protocols, signs and symptoms of early and intermediate complications such as infection, GvHD and late onset VOD as well as others.

The nurse plays an important part in navigating follow-up visits, is often involved in arranging transfusion support and offers medication advice, information and replenishment. They reinforce the safety messages given to the patient at discharge particularly with regard to infections, signs and symptoms to report and the risks of delays in this.

The nurse working in the area of posttransplant care is also well placed to offer practical guidance on food and nutrition, exercise, relationships and sex. These topics among many others are not always easily broached in the transplant clinic and the nurse–patient relationship supports this dialogue. Conversations around the emotional toil of transplant recovery are important to identify those for whom more formal psychological support may be beneficial and the use of standardised assessment tools validate a range of concerns, triggering conversations that enable referrals to other services as needed.

Nurses have become increasingly involved in late effects care. The clear guidelines for late effects surveillance and screening have been incorporated within local SOPs, facilitating the development of nurse roles in this area. Support with health behaviour and lifestyle modifications such as smoking cessation, exercise and nutrition; and also, immunisation, sexual function and vocational rehabilitation are components of late effects care amenable to the attention of nurses working in the long-term follow-up clinic.

32.8 Ambulatory and Day Unit Nursing

Ambulatory and day unit nursing is an evolving area. Treatments that were previously given in the in-patient setting are now transferring to ambulatory care and day units, such as complex cytotoxic and supportive medications. The result is that the nursing staff in these areas now require skills on par to colleagues working on in-patient departments. Many hospitals now operate a rotation of staff through all patient areas. Nurses in ambulatory and day care require additional skills such as management of a computerised ambulatory delivery device or CADD pumps and how to infuse DLI that are different from ward-based teams.

These areas are usually small requiring fewer staff with the expectation that the patient population is pre-selected and not in need of routine medical review. Nurses will need to be competent in managing acutely unwell patients and be aware of how to access hot beds and medical team assessments out of hours. There should be clear pathways for rapid hospital assessment and treatment of potential sepsis. This exciting area offers nurses opportunities to develop their roles and ultimately improve patient care.

32.9 Summary

It is clear that nurses working in HCT have a wealth of opportunities to make a difference to patients with each role described here offering a vital component of care. Nurses are a vital component of the transplant team complementing the other roles in the MDT. Continuous modifications and developments in treatment undoubtedly influence this highly complex but highly rewarding area of nursing care.

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